A constructive lower bound for primes $3 \pmod 4$

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Abstract. We give a Euclidean construction of infinitely many primes $3 \pmod 4$ and obtain an explicit, albeit weak, lower bound on their number up to x. The argument uses only elementary factorization and congruences.

Sequence and notation. OEIS A002145 consists of the primes $3, 7, 11, 19, 23, \ldots$ congruent to $3 \pmod 4$. Let $\pi_{3 \pmod 4}(x)$ count such primes at most x.

Theorem 1. *For every real $x \geq 4$,*

$$\pi_{3 \pmod 4}(x) \geq \lfloor \log_2 (1 + \log_4 x) \rfloor .$$

In particular, OEIS A002145 contains infinitely many distinct primes.

Proof. Set $P_0 = 1$. Suppose we have chosen pairwise distinct primes $p_1, \ldots, p_m \equiv 3 \pmod 4$ and put $P_m = \prod_{j=1}^{m} p_j$. The integer

$$N_m = 4P_m - 1 > 1$$

is odd and $N_m \equiv 3 \pmod 4$. Any odd integer $3 \pmod 4$ has an odd number of prime factors $3 \pmod 4$, counted with multiplicity: in its prime factorization, the factors congruent to $1 \pmod 4$ do not change the residue, whereas each factor $3 \pmod 4$ reverses it. Therefore N_m has at least one prime divisor $p_{m+1} \equiv 3 \pmod 4$. For each $j \leq m$, since $p_j \mid P_m$,

$$N_m \equiv -1 \pmod{p_j},$$

so p_{m+1} is distinct from all preceding primes.

The product of the first $m+1$ chosen primes satisfies

$$P_{m+1} = P_m p_{m+1} \leq P_m(4P_m - 1) < 4P_m^2.$$

Induction from $P_0 = 1$ therefore yields

$$P_m \leq 4^{2^m - 1} \quad (m \geq 0).$$

Indeed, $4(4^{2^m-1})^2 = 4^{2^{m+1}-1}$. For any integer $m \geq 0$ with $2^m \leq 1 + \log_4 x$ we get $P_m \leq x$; each of the m distinct primes p_j is at most P_m. The greatest such integer m is $\lfloor \log_2(1 + \log_4 x) \rfloor$, giving the asserted inequality. □

Scope and priority. The infinitude argument is classical, and the displayed explicit bound is an elementary corollary of this construction. The note claims neither novelty nor a solution to an open problem such as the twin-prime conjecture.

References.

1. N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, A002145, "Primes of the form $4k + 3$," consulted 12 November 2026.
2. G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008; classical elementary Euclid-style infinitude arguments.

Primary author (AI): ChatGPT (OpenAI). This note was produced with AI assistance. The argument was checked by independent algebraic derivation and computational spot checks; it has not been independently refereed. All statements and errors remain subject to human verification. This note makes no claim of novelty.